MATH1101: Differential Calculus & Analytical Geometry

Complete Exam Notes · 1st Year 1st Semester · PUST · Module 2 (Units 4 & 5): Differentiability, Successive Differentiation & Leibniz's Theorem

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Module 2 Overview

Units Covered in Module 2:

- **Unit 4:** Concept of Differentiability (L.H.D. & R.H.D.), Continuity vs. Differentiability, Geometric Interpretation of Derivative (Secant & Tangent Lines, Corner Points, Cusps, Vertical Tangents), and Absolute Maxima & Minima on Closed Intervals.
- **Unit 5:** Successive Differentiation (n-th Derivative Formulas y_n, Partial Fractions & Trigonometric Reduction) and Leibniz's Theorem (Proof by Induction, y_{n+2} Differential Equations & Evaluating $y_n(0)$).

Master Roadmap (MATH 1101)

Mod.	Units & Core Syllabus Topics	Main Sources	Status
1	Units 1–3: Functions, Domain/Range, Graphs, Inverse, Limits & Continuity	Class Notes · Anton Ch. 0, 1 · PUST 20–23	Module 1
2	Units 4–5: Differentiability, Geometric Meaning, Extrema, y_n & Leibniz	Class Notes · Anton Ch. 2, 4 · PUST 20–23	✓ **This file**
3	Units 6–9: Rolle's & MVT, Taylor/Maclaurin, L'Hôpital & Curve Tracing	Class Notes · Anton Ch. 3, 4, 9 · Handout	Module 3
4	Units 10–13: Tangents, Asymptotes, Curvature, Partial Deriv., Euler & Jacobian	Akib Notes · Anton Ch. 13 · PUST 20–23	Module 4
5	Units 14–15: 2D Coordinates, Transformation of Axes & Pair of Straight Lines	Anton Ch. 10 · Rahman & Bhattacharjee	Module 5
6	Units 16–18: Conic Reduction, Parabola, Ellipse, Hyperbola & System of Circles	Anton Ch. 10 · Rahman & Bhattacharjee	Module 6
7	Units 19–21: 3D Coordinates, DCs/DRs, Projections, Planes & Skew Lines (S.D.)	Anton Ch. 11 · Rahman & Bhattacharjee	Module 7
8	Units 22–24: Sphere, Cone, Right Circular Cylinder & 3D Central Conicoids	Anton Ch. 11 · Rahman & Bhattacharjee	Module 8

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1 Unit 4: Differentiability, Geometric Interpretation & Absolute Extrema

1.1 Definition of Derivative and Differentiability at a Point

Definition (First-Principle Derivative): The derivative of $y = f(x)$ with respect to x is defined as:

$$\frac{dy}{dx} = f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

provided this limit exists. At $x = a$, the one-sided derivatives are:

$$\text{Left-Hand Derivative (L.H.D.):} \quad Lf'(a) = \lim_{h\to 0^-} \frac{f(a+h) - f(a)}{h} = \lim_{x\to a^-} \frac{f(x) - f(a)}{x - a},$$

$$\text{Right-Hand Derivative (R.H.D.):} \quad Rf'(a) = \lim_{h\to 0^+} \frac{f(a+h) - f(a)}{h} = \lim_{x\to a^+} \frac{f(x) - f(a)}{x - a}.$$

Condition for Differentiability: $f(x)$ is differentiable at $x = a$ if and only if L.H.D. = R.H.D. (both finite).

1.2 Relationship Between Differentiability and Continuity

Theorem: *Differentiability implies continuity, but the converse is not true.*

Proof: Assume $f(x)$ is differentiable at $x = a$, so $f'(a) = \lim_{x\to a} \frac{f(x) - f(a)}{x - a}$ exists finitely. Then for $x \neq a$:

$$\begin{aligned}
\lim_{x\to a}\big[f(x) - f(a)\big] &= \lim_{x\to a}\left[\frac{f(x) - f(a)}{x - a} \cdot (x - a)\right] \\
&= \left(\lim_{x\to a} \frac{f(x) - f(a)}{x - a}\right) \cdot \Big(\lim_{x\to a}(x - a)\Big) \\
&= f'(a) \cdot 0 = 0 \\
&\implies \lim_{x\to a} f(x) = f(a).
\end{aligned}$$

Hence $f(x)$ is continuous at $x = a$. ■

> **Class Lecture & PUST EECE 2022–23 Exam Q1(c) [5.5 Marks]**
> **Question:** Prove that the function $f(x) = |x|$ is continuous at $x = 0$, but not differentiable at $x = 0$.

Solution (Exam Style): By definition of the modulus function:

$$f(x) = |x| = \begin{cases} -x, & x < 0 \\ 0, & x = 0 \\ x, & x > 0 \end{cases}$$

Step 1 (Continuity Test at $x = 0$):

$$\begin{aligned}
&\text{At } x = 0, \quad f(0) = 0 \\
&\text{L.H.L.} = \lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} (-x) = 0 \\
&\text{R.H.L.} = \lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} (x) = 0
\end{aligned}$$

Since L.H.L. = R.H.L. = $f(0) = 0$, $f(x) = |x|$ is **continuous at** $x = 0$.

Step 2 (Differentiability Test at $x = 0$):

$$\begin{aligned}
\text{L.H.D.} &= \lim_{h\to 0^-} \frac{f(0+h) - f(0)}{h} \\
&= \lim_{h\to 0^-} \frac{|h| - 0}{h} \\
&= \lim_{h\to 0^-} \frac{-h}{h} = -1 \quad [\because \text{for } h < 0,\ |h| = -h]
\end{aligned}$$

$$\text{R.H.D.} = \lim_{h\to 0^+} \frac{f(0+h)-f(0)}{h}$$
$$= \lim_{h\to 0^+} \frac{|h|-0}{h}$$
$$= \lim_{h\to 0^+} \frac{+h}{h} = +1 \quad [\because \text{for } h>0,\ |h| = +h]$$

Since L.H.D. $\neq$ R.H.D. $(-1 \neq +1)$, $f'(0)$ does not exist. Therefore, $f(x) = |x|$ is **not differentiable at** $x = 0$. ■

PUST EECE 2021–22 Semester Exam — Question 2(a) [5 Marks]

Question: Discuss the continuity and differentiability of $f(x)$ at $x = 1$, where

$$f(x) = \begin{cases} 2x, & x < 0 \\ 2x^2, & 0 \le x < 1 \\ 4 - 2x, & x \ge 1 \end{cases}$$

Solution (Exam Style):

Step 1 (Continuity at $x = 1$):

$$f(1) = 4 - 2(1) = 2$$
$$\text{L.H.L.} = \lim_{x\to 1^-} (2x^2) = 2(1)^2 = 2$$
$$\text{R.H.L.} = \lim_{x\to 1^+} (4 - 2x) = 4 - 2(1) = 2$$

Since L.H.L. = R.H.L. = $f(1) = 2$, $f(x)$ is **continuous at** $x = 1$.

Step 2 (Differentiability at $x = 1$):

$$\text{L.H.D.} = \lim_{h\to 0^-} \frac{f(1+h)-f(1)}{h}$$
$$= \lim_{h\to 0^-} \frac{2(1+h)^2 - 2}{h} \quad [\text{For } x<1,\ f(x) = 2x^2]$$
$$= \lim_{h\to 0^-} \frac{4h + 2h^2}{h} = \lim_{h\to 0^-} (4 + 2h) = 4$$
$$\text{R.H.D.} = \lim_{h\to 0^+} \frac{f(1+h)-f(1)}{h}$$
$$= \lim_{h\to 0^+} \frac{[4 - 2(1+h)] - 2}{h} \quad [\text{For } x \ge 1,\ f(x) = 4 - 2x]$$
$$= \lim_{h\to 0^+} \frac{-2h}{h} = -2$$

Since L.H.D. $= 4 \neq$ R.H.D. $= -2$, $f(x)$ is **not differentiable at** $x = 1$.

Problem 4.1 (Class Lecture): Test continuity and differentiability of $f(x) = \begin{cases} x^2 - 1, & x \le 1 \\ 2x - 2, & x > 1 \end{cases}$ **at** $x = 1$.

Solution (Exam Style):

Continuity at $x = 1$: $\text{L.H.L.} = \lim_{x\to 1^-} (x^2 - 1) = 0$, $\text{R.H.L.} = \lim_{x\to 1^+} (2x - 2) = 0$, $f(1) = 0$.

Differentiability at $x = 1$: $\text{L.H.D.} = \lim_{h\to 0^-} \frac{[(1+h)^2 - 1] - 0}{h} = \lim_{h\to 0^-} (2 + h) = 2$,

$\text{R.H.D.} = \lim_{h\to 0^+} \frac{[2(1+h) - 2] - 0}{h} = \lim_{h\to 0^+} \frac{2h}{h} = 2.$

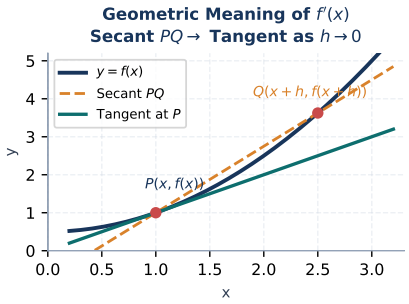
Since L.H.D. = R.H.D. = 2, $f(x)$ is both **continuous and differentiable at** $x = 1$, and $f'(1) = 2$.

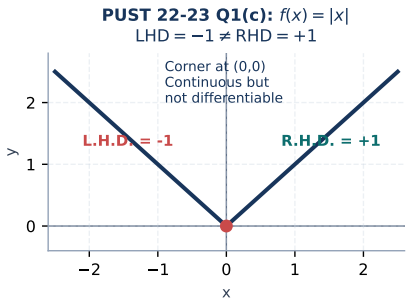
1.3 Geometric Interpretation of Derivative & Non-Differentiable Points

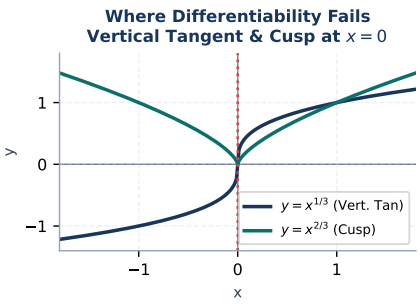
- **Slope of Tangent Line:** The ratio $\dfrac{f(x+h)-f(x)}{h}$ is the slope of the **secant line** through $P(x, f(x))$ and $Q(x+h, f(x+h))$. As $h \to 0$, $Q \to P$ and the secant line approaches the **tangent line** at P with slope:

$$m_{\tan} = \tan\theta = f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}.$$

- **Three Geometric Cases Where a Continuous Function Fails to Be Differentiable at $x = a$:**
  1. **Corner Point:** L.H.D. and R.H.D. are both finite, but L.H.D. $\neq$ R.H.D. (e.g., $f(x) = |x|$ at $x = 0$).
  2. **Vertical Tangent Line:** $f(x)$ is continuous at $x = a$ and $\lim_{x\to a^-} f'(x) = \lim_{x\to a^+} f'(x) = +\infty$ or $-\infty$ (e.g., $f(x) = x^{1/3}$ at $x = 0$).
  3. **Cusp Point:** $\lim_{x\to a} f'(x) = +\infty$ from one side and $-\infty$ from the other side (e.g., $f(x) = x^{2/3}$ at $x = 0$).







1.4 Absolute (Global) Maxima and Minima on a Closed Interval

Working Rule (Closed-Interval Method): To find the absolute maximum and minimum of a continuous function $f(x)$ on $[a, b]$:

1. Find all critical points $c \in (a, b)$ where $f'(c) = 0$ or $f'(c)$ does not exist.
2. Evaluate $f(x)$ at each critical point in (a, b) and at the endpoints $x = a$ and $x = b$.
3. The largest value is the **absolute maximum** and the smallest value is the **absolute minimum**.

Problem 4.2 (Anton Sec. 4.4): Find the absolute extrema of $f(x) = 2x^3 - 15x^2 + 36x$ on $[1, 5]$.

Solution (Exam Style):

$f'(x) = 6x^2 - 30x + 36 = 6(x-2)(x-3)$
For critical points, $f'(x) = 0 \implies x = 2,\ 3 \in (1, 5)$.

Evaluating $f(x)$ at the critical points and endpoints:

$f(1) = 2(1)^3 - 15(1)^2 + 36(1) = 23$ [Smallest value $\implies$ Absolute Minimum]
$f(2) = 2(8) - 15(4) + 36(2) = 28$
$f(3) = 2(27) - 15(9) + 36(3) = 27$
$f(5) = 2(125) - 15(25) + 36(5) = 55$ [Largest value $\implies$ Absolute Maximum]

$\therefore$ Absolute Maximum $= 55$ (at $x = 5$) and Absolute Minimum $= 23$ (at $x = 1$).

No.	Function $y = f(x)$	n-th Derivative Formula $y_n = \dfrac{d^n y}{dx^n}$
1	$y = x^m$	$y_n = m(m-1)\cdots(m-n+1)\,x^{m-n}$. If $m \in \mathbb{N}$: $D^m(x^m) = m!$, and $D^n(x^m) = 0$ for $n > m$.
2	$y = (ax+b)^m$	$y_n = m(m-1)\cdots(m-n+1)\,a^n(ax+b)^{m-n}$.
3	$y = \dfrac{1}{ax+b}$	$y_n = \dfrac{(-1)^n\,n!\,a^n}{(ax+b)^{n+1}}$. Special case: $D^n\left(\dfrac{1}{x}\right) = \dfrac{(-1)^n\,n!}{x^{n+1}}$.
4	$y = \ln(ax+b)$	$y_n = \dfrac{(-1)^{n-1}\,(n-1)!\,a^n}{(ax+b)^n}$.
5	$y = e^{mx}$ / $y = a^{mx}$	$D^n(e^{mx}) = m^n e^{mx}$; $D^n(a^{mx}) = m^n(\ln a)^n a^{mx}$.
6	$y = \sin(ax+b)$	$y_n = a^n \sin\left(ax+b+\dfrac{n\pi}{2}\right)$.
7	$y = \cos(ax+b)$	$y_n = a^n \cos\left(ax+b+\dfrac{n\pi}{2}\right)$.
8	$y = e^{ax}\sin(bx+c)$	$y_n = (a^2+b^2)^{n/2}e^{ax}\sin\big(bx+c+n\tan^{-1}(b/a)\big)$.
9	$y = e^{ax}\cos(bx+c)$	$y_n = (a^2+b^2)^{n/2}e^{ax}\cos\big(bx+c+n\tan^{-1}(b/a)\big)$.

2 Unit 5: Successive Differentiation and Leibniz's Theorem

2.1 Standard n-th Derivative Formulas ($y_n = D^n y$)

Problem 5.1 (Class Lecture): Find y_n if $y = \dfrac{x}{(x-1)(x-2)(x-3)}$.

Solution (Exam Style): Let the partial fraction decomposition be:

$$\frac{x}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$$

Finding A, B, C by the cover-up rule:

$$\text{For } x=1: \quad A = \frac{1}{(1-2)(1-3)} = \frac{1}{(-1)(-2)} = \frac{1}{2}$$

$$\text{For } x=2: \quad B = \frac{2}{(2-1)(2-3)} = \frac{2}{(1)(-1)} = -2$$

$$\text{For } x=3: \quad C = \frac{3}{(3-1)(3-2)} = \frac{3}{(2)(1)} = \frac{3}{2}$$

Substituting A, B, C and differentiating n times using $D^n\left(\frac{1}{x-k}\right) = \frac{(-1)^n n!}{(x-k)^{n+1}}$:

$$y = \frac{1}{2(x-1)} - \frac{2}{x-2} + \frac{3}{2(x-3)}$$

$$\implies y_n = (-1)^n n! \left[\frac{1}{2(x-1)^{n+1}} - \frac{2}{(x-2)^{n+1}} + \frac{3}{2(x-3)^{n+1}}\right].$$

Problem 5.2 (Class Lecture): Find y_n for (a) $y = \sin^3 x$ and (b) $y = \sin x \sin 3x$.

Solution (Exam Style):

(a) Given, $y = \sin^3 x$

$$= \frac{1}{4}\big(3\sin x - \sin 3x\big) \quad [\because \sin 3x = 3\sin x - 4\sin^3 x]$$

$$\implies y_n = \frac{3}{4}\sin\left(x + \frac{n\pi}{2}\right) - \frac{3^n}{4}\sin\left(3x + \frac{n\pi}{2}\right).$$

(b) Given, $y = \sin x \sin 3x = \dfrac{1}{2}\big(2\sin 3x \sin x\big)$

$$= \frac{1}{2}\big(\cos 2x - \cos 4x\big) \quad [\because 2\sin A \sin B = \cos(A-B) - \cos(A+B)]$$

$$\implies y_n = \frac{1}{2}\left[2^n \cos\left(2x + \frac{n\pi}{2}\right) - 4^n \cos\left(4x + \frac{n\pi}{2}\right)\right].$$

2.2 Leibniz's Theorem: Statement, Proof & Solved Exam Problems

Leibniz's Theorem (Class Lecture & PUST EECE 2022–23 Exam Q2b) [4.5 Marks]
Statement: If u and v are two functions of x possessing derivatives up to the n-th order, then the n-th derivative of their product $y = uv$ is given by:

$$(uv)_n = u_n v + \binom{n}{1} u_{n-1} v_1 + \binom{n}{2} u_{n-2} v_2 + \cdots + \binom{n}{r} u_{n-r} v_r + \cdots + u v_n.$$

Proof by Mathematical Induction (Exam Style):

1. **Base Step ($n = 1, 2$):** By direct differentiation of uv:

$$(uv)_1 = u_1 v + u v_1 = u_1 v + \binom{1}{1} u v_1,$$
$$(uv)_2 = u_2 v + 2u_1 v_1 + u v_2 = u_2 v + \binom{2}{1} u_1 v_1 + \binom{2}{2} u v_2.$$

   Thus the theorem holds for $n = 1$ and $n = 2$.

2. **Inductive Hypothesis:** Assume the formula holds for $n = m$:

$$(uv)_m = u_m v + \binom{m}{1} u_{m-1} v_1 + \binom{m}{2} u_{m-2} v_2 + \cdots + \binom{m}{r} u_{m-r} v_r + \cdots + u v_m.$$

3. **Inductive Step ($n = m + 1$):** Differentiating both sides once more with respect to x:

$$\begin{aligned}(uv)_{m+1} &= (u_{m+1} v + u_m v_1) + \binom{m}{1}(u_m v_1 + u_{m-1} v_2) + \cdots + (u_1 v_m + u v_{m+1}) \\ &= u_{m+1} v + \left[\binom{m}{0} + \binom{m}{1}\right] u_m v_1 + \cdots + \left[\binom{m}{r-1} + \binom{m}{r}\right] u_{m+1-r} v_r + \cdots + u v_{m+1} \\ &= u_{m+1} v + \binom{m+1}{1} u_m v_1 + \cdots + \binom{m+1}{r} u_{m+1-r} v_r + \cdots + u v_{m+1} \quad \left[\because \binom{m}{r-1} + \binom{m}{r} = \binom{m+1}{r}\right].\end{aligned}$$

   Hence the theorem holds for $n = m + 1$, and by mathematical induction for all $n \in \mathbb{N}$. ■

Problem 5.3 (Class Lecture): If $y = a\cos(\ln x) + b\sin(\ln x)$, show that $x^2 y_{n+2} + (2n+1) x y_{n+1} + (n^2+1) y_n = 0$.

Solution (Exam Style):

Given, $y = a\cos(\ln x) + b\sin(\ln x)$

$$\begin{aligned}&\implies y_1 = \frac{-a\sin(\ln x) + b\cos(\ln x)}{x} \quad \text{[Differentiating w.r.t. } x\text{]} \\ &\implies x y_1 = -a\sin(\ln x) + b\cos(\ln x) \\ &\implies x y_2 + y_1 = \frac{-a\cos(\ln x) - b\sin(\ln x)}{x} \quad \text{[Differentiating again w.r.t. } x\text{]} \\ &\implies x^2 y_2 + x y_1 = -\left[a\cos(\ln x) + b\sin(\ln x)\right] = -y \\ &\implies x^2 y_2 + x y_1 + y = 0\end{aligned}$$

Differentiating n times by Leibniz's Theorem:

$$\begin{aligned}&\left[x^2 y_{n+2} + 2nx y_{n+1} + n(n-1) y_n\right] + \left[x y_{n+1} + n y_n\right] + y_n = 0 \\ &\implies x^2 y_{n+2} + (2n+1) x y_{n+1} + (n^2+1) y_n = 0. \quad ■\end{aligned}$$

Class Lecture & PUST EECE 2020–21 Exam Q2(a) [5 Marks]
Question: If $y = \sin(m \sin^{-1} x)$, show that $(1-x^2) y_{n+2} - (2n+1) x y_{n+1} + (m^2 - n^2) y_n = 0$. Also find $y_n(0)$.

Solution (Exam Style):

Given, $y = \sin(m \sin^{-1} x)$

$$\implies y_1 = \frac{m\cos(m\sin^{-1}x)}{\sqrt{1-x^2}}$$
$$\implies (1-x^2)y_1^2 = m^2\cos^2(m\sin^{-1}x) \quad \text{[Cross-multiplying and squaring]}$$
$$\implies (1-x^2)y_1^2 = m^2(1-y^2) \quad [\because \cos^2\theta = 1-\sin^2\theta = 1-y^2]$$
$$\implies (1-x^2)(2y_1y_2) - 2xy_1^2 = -2m^2yy_1 \quad \text{[Differentiating w.r.t. } x\text{]}$$
$$\implies (1-x^2)y_2 - xy_1 + m^2y = 0 \quad \text{[Dividing by } 2y_1 \neq 0\text{]} \tag{1}$$

Applying Leibniz's Theorem to differentiate (1) n times:

$$\big[(1-x^2)y_{n+2} - 2nxy_{n+1} - n(n-1)y_n\big] - \big[xy_{n+1} + ny_n\big] + m^2y_n = 0$$
$$\implies (1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2-n^2)y_n = 0. \tag{2}$$

Finding $y_n(0)$: Putting $x = 0$ in y, y_1, (1), and (2):

$$y(0) = 0, \qquad y_1(0) = m, \qquad y_2(0) = 0,$$
$$y_{n+2}(0) = (n^2-m^2)y_n(0) \quad \text{[Recurrence relation at } x = 0\text{]}.$$

- **When n is even:** $y_n(0) = 0$.
- **When n is odd ($n \geq 3$):** $y_n(0) = m(1^2-m^2)(3^2-m^2)(5^2-m^2)\cdots\big((n-2)^2-m^2\big)$. ■

PUST EECE 2022–23 Exam Q4(c) [5 Marks] & 2021–22 Exam Q2(b) [5 Marks]

Problem A (2022–23 Q4c): If $y = e^{a\sin^{-1}x}$, show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0$.
Solution (Exam Style):

$y = e^{a\sin^{-1}x}$

$$\implies y_1 = \frac{ae^{a\sin^{-1}x}}{\sqrt{1-x^2}} = \frac{ay}{\sqrt{1-x^2}}$$
$$\implies (1-x^2)y_1^2 = a^2y^2$$
$$\implies (1-x^2)(2y_1y_2) - 2xy_1^2 = 2a^2yy_1$$
$$\implies (1-x^2)y_2 - xy_1 - a^2y = 0 \quad \text{[Dividing by } 2y_1 \neq 0\text{]}$$
$$\implies (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0 \quad \text{[By Leibniz's Theorem]}.$$ ■

Problem B (2021–22 Q2b): If $y = (x^2-1)^n$, prove that $(x^2-1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0$.
Solution (Exam Style):

$y = (x^2-1)^n$

$$\implies y_1 = 2nx(x^2-1)^{n-1}$$
$$\implies (x^2-1)y_1 = 2nxy$$
$$\implies (x^2-1)y_2 + 2xy_1 = 2n(xy_1+y) \quad \text{[Differentiating w.r.t. } x\text{]}$$
$$\implies (x^2-1)y_2 - 2(n-1)xy_1 - 2ny = 0$$
$$\implies \big[(x^2-1)y_{n+2} + 2nxy_{n+1} + n(n-1)y_n\big] - 2(n-1)\big[xy_{n+1} + ny_n\big] - 2ny_n = 0 \quad \text{[By Leibniz's Theorem]}$$
$$\implies (x^2-1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0.$$ ■

Problem 5.4 (Class Lecture): If $y^{1/m} + y^{-1/m} = 2x$, show that $(x^2-1)y_{n+2} + (2n+1)xy_{n+1} + (n^2-m^2)y_n = 0$.

Solution (Exam Style):

$$y^{1/m} + \frac{1}{y^{1/m}} = 2x$$

$$\implies \left(y^{1/m}\right)^2 - 2x\left(y^{1/m}\right) + 1 = 0$$
$$\implies y^{1/m} = x \pm \sqrt{x^2-1}$$
$$\implies y = \left(x \pm \sqrt{x^2-1}\right)^m$$
$$\implies y_1 = \frac{\pm my}{\sqrt{x^2-1}}$$
$$\implies (x^2-1)y_1^2 = m^2y^2$$
$$\implies (x^2-1)y_2 + xy_1 - m^2y = 0 \quad \text{[Differentiating and dividing by } 2y_1\text{]}$$
$$\implies (x^2-1)y_{n+2} + (2n+1)xy_{n+1} + (n^2-m^2)y_n = 0 \quad \text{[By Leibniz's Theorem]}. \quad \blacksquare$$

2.3 Past-Question Map for Module 2

Session	Questions Asked from Module 2 (Units 4–5)	Marks
2022–23	Q1(c) Prove $f(x) = \|x\|$ is continuous but not differentiable at $x = 0$ · Q2(b) State & prove Leibniz's theorem · Q4(c) $y = e^{a\sin^{-1}x}$	$5.5 + 4.5 + 5$
2021–22	Q2(a) Continuity & differentiability of piecewise $f(x)$ at $x = 1$ · Q2(b) Leibniz's theorem for $y = (x^2-1)^n$	$5 + 5$
2020–21	Q2(a) If $y = \sin(m\sin^{-1}x)$, prove $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2-n^2)y_n = 0$ and find $y_n(0)$	5

Quick Revision — Module 2 (Units 4–5)

• **Differentiability:** $Lf'(a) = \lim_{h\to 0^-}\frac{f(a+h)-f(a)}{h} = Rf'(a)$; Differentiable $\implies$ Continuous (converse false: $|x|$ at 0).
• **Key y_n Formulas:** $D^n\left(\frac{1}{ax+b}\right) = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$; $D^n\sin(ax+b) = a^n\sin\left(ax+b+\frac{n\pi}{2}\right)$; $D^n\cos(ax+b) = a^n\cos\left(ax+b+\frac{n\pi}{2}\right)$.
• **Leibniz Shortcuts:** $D^n[(1-x^2)y_2] = (1-x^2)y_{n+2} - 2nxy_{n+1} - n(n-1)y_n$; $D^n[xy_1] = xy_{n+1} + ny_n$.

Common Mistakes to Avoid in the Exam

• Differentiating a quotient with $\sqrt{1-x^2}$ in the denominator directly: always cross-multiply $\sqrt{1-x^2}\,y_1$ and square both sides before finding y_2!
• Forgetting to check whether n is odd or even when evaluating $y_n(0)$ from the recurrence $y_{n+2}(0) = (n^2-m^2)y_n(0)$.